

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Iron(III) fluoride trihydrate

**CAS No** 15469-38-2

**Synonyms** Ferric Fluoride Trihydrate

**Product Code** **13068**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

|                                             |              |
|---------------------------------------------|--------------|
| Acute Oral Toxicity                         | Category 4   |
| Acute Dermal Toxicity                       | Category 4   |
| Acute Inhalation Toxicity - Dusts and Mists | Category 4   |
| Skin Corrosion/Irritation                   | Category 1 B |
| Serious Eye Damage/Eye Irritation           | Category 1   |

#### Environmental hazards

No hazards identified

#### Label Elements



Exclamation Mark



Corrosion

**Signal Word****Danger****Hazard Statements**

H314 - Causes severe skin burns and eye damage

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P331 - Do NOT induce vomiting

P330 - Rinse mouth

P363 - Wash contaminated clothing before reuse

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

No information available

## Section 3 - Composition and Information on Ingredients

| Component                     | CAS No     | Weight % |
|-------------------------------|------------|----------|
| Iron(III) fluoride trihydrate | 15469-38-2 | <=100    |
| Ferric fluoride               | 7783-50-8  | -        |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Ingestion**

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | No special precautions required.                                                                                                                                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Not combustible. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Hydrogen fluoride, Metal oxides.

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

### Environmental Precautions

Should not be released into the environment. Do not allow material to contaminate ground water system. See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

**Conditions for Safe Storage, Including any Incompatibilities**

Store under an inert atmosphere. Protect from moisture. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

| Component       | Australia                                           | New Zealand WEL | ACGIH TLV                                           | The United Kingdom                                                | Germany                                                                                                            |
|-----------------|-----------------------------------------------------|-----------------|-----------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Ferric fluoride | TWA: 1 mg/m <sup>3</sup> TWA: 2.5 mg/m <sup>3</sup> |                 | TWA: 1 mg/m <sup>3</sup> TWA: 2.5 mg/m <sup>3</sup> | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr | TWA: 1 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 4<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). MAK<br>Haut |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

None under normal use conditions. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |                 |                       |
| Neoprene       |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particle filter (or AUS/NZ equivalent)

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                         |                                     |                                   |
|-----------------------------------------|-------------------------------------|-----------------------------------|
| Appearance                              | Off-white - Pale brown              |                                   |
| Physical State                          | Solid Crystalline                   |                                   |
| Odor                                    | No information available            |                                   |
| Odor Threshold                          | No data available                   |                                   |
| pH                                      | No information available            |                                   |
| Melting Point/Range                     | No data available                   |                                   |
| Softening Point                         | No data available                   |                                   |
| Boiling Point/Range                     | No information available            |                                   |
| Flash Point                             | No information available            | Method - No information available |
| Evaporation Rate                        | Not applicable                      | Solid                             |
| Flammability (solid,gas)                | No information available            |                                   |
| Explosion Limits                        | No data available                   |                                   |
| Vapor Pressure                          | No data available                   |                                   |
| Vapor Density                           | Not applicable                      | Solid                             |
| Specific Gravity / Density              | No data available                   |                                   |
| Bulk Density                            | No data available                   |                                   |
| Water Solubility                        | No information available            |                                   |
| Solubility in other solvents            | No information available            |                                   |
| Partition Coefficient (n-octanol/water) |                                     |                                   |
| Autoignition Temperature                | No data available                   |                                   |
| Decomposition Temperature               | No data available                   |                                   |
| Viscosity                               | Not applicable                      | Solid                             |
| Explosive Properties                    | No information available            |                                   |
| Oxidizing Properties                    | No information available            |                                   |
| Other information                       |                                     |                                   |
| Molecular Formula                       | FeF <sub>3</sub> .3H <sub>2</sub> O |                                   |
| Molecular Weight                        | 166.89 (112.84anhy)                 |                                   |

## Section 10 - Stability and Reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactivity                       | None known, based on information available |
| Stability                        | Hygroscopic.                               |
| Conditions to Avoid              | Exposure to moist air or water.            |
| Incompatible Materials           | Oxidizing agent.                           |
| Hazardous Decomposition Products | Hydrogen fluoride. Metal oxides.           |
| Hazardous Polymerization         | No information available.                  |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

### Product Information

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) acute toxicity;                        |                                                                                                                                                                                                                                                            |
| Oral                                       | Category 4<br>Data from closely analogous substances                                                                                                                                                                                                       |
| Dermal                                     | Category 4<br>Data from closely analogous substances                                                                                                                                                                                                       |
| Inhalation                                 | Category 4<br>Data from closely analogous substances                                                                                                                                                                                                       |
| (b) skin corrosion/irritation;             | Category 1 B                                                                                                                                                                                                                                               |
| (c) serious eye damage/irritation;         | Category 1                                                                                                                                                                                                                                                 |
| (d) respiratory or skin sensitization;     |                                                                                                                                                                                                                                                            |
| Respiratory                                | No data available                                                                                                                                                                                                                                          |
| Skin                                       | No data available                                                                                                                                                                                                                                          |
| (e) germ cell mutagenicity;                | No data available                                                                                                                                                                                                                                          |
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                                                                                                                                                                             |
| (g) reproductive toxicity;                 | No data available                                                                                                                                                                                                                                          |
| (h) STOT-single exposure;                  | No data available                                                                                                                                                                                                                                          |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                          |
| Target Organs                              | No information available.                                                                                                                                                                                                                                  |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                                                                                                                                                                    |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

## Section 12 - Ecological Information

|                                 |                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects             | Do not allow material to contaminate ground water system.                                                         |
| Persistence and Degradability   | Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary |
| Persistence                     | Persistence is unlikely.                                                                                          |
| Degradability                   | Not relevant for inorganic substances.                                                                            |
| Bioaccumulative Potential       | Bioaccumulation is unlikely                                                                                       |
| Mobility                        | No information available.                                                                                         |
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors                                         |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance                                                    |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance                                                    |

## Section 13 - Disposal Considerations

|                            |                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------|
| Waste from Residues/Unused | Do not allow into drains or watercourses or dispose of where ground or surface waters may |
|----------------------------|-------------------------------------------------------------------------------------------|

|                               |                                                                                                                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Products</b>               | be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                               |
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                |
| <b>Other Information</b>      | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                            |
|--------------------------------|--------------------------------------------|
| <b>UN-No</b>                   | UN3260                                     |
| <b>Proper Shipping Name</b>    | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Technical Shipping Name</b> | (Iron(III) fluoride trihydrate)            |
| <b>Hazard Class</b>            | 8                                          |
| <b>Packing Group</b>           | II                                         |

### ADG

|                                |                                            |
|--------------------------------|--------------------------------------------|
| <b>UN-No</b>                   | UN3260                                     |
| <b>Proper Shipping Name</b>    | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Technical Shipping Name</b> | (Iron(III) fluoride trihydrate)            |
| <b>Hazard Class</b>            | 8                                          |
| <b>Packing Group</b>           | II                                         |

### IATA

|                                |                                            |
|--------------------------------|--------------------------------------------|
| <b>UN-No</b>                   | UN3260                                     |
| <b>Proper Shipping Name</b>    | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Technical Shipping Name</b> | (Iron(III) fluoride trihydrate)            |
| <b>Hazard Class</b>            | 8                                          |
| <b>Packing Group</b>           | II                                         |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations                      **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                                  | Standard for the Uniform Scheduling of Medicines and Poisons                      |
|--------------------------------------------|-----------------------------------------------------------------------------------|
| Iron(III) fluoride trihydrate - 15469-38-2 | Schedule 2 listed<br>Schedule 4 listed - in injectable preparations for human use |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient; in preparations for injection except in preparations containing ≤0.1% of Iron</p> <p>Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient; in other preparations except in liquid or gel preparations containing ≤0.1% of Iron, or in animal feeds or feed premixes</p> <p>Schedule 5 listed - for use as agricultural chemicals except in preparations containing ≤4% of Iron</p> <p>Schedule 6 listed - except up to 1% of Iron oxides when present as an excipient. For the treatment of animals except: when included in Schedule 5, in liquid or gel preparations containing ≤0.1% of Iron, or in animal feeds or feed premixes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Ferric fluoride - 7783-50-8 | <p>Schedule 2 listed</p> <p>Schedule 3 listed</p> <p>Schedule 4 listed - in preparations for human use except when included in or expressly excluded from Schedule 2 or 3</p> <p>Schedule 4 listed - in injectable preparations for human use</p> <p>Schedule 5 listed - in preparations except: in preparations for human use, or in preparations containing ≤15 mg/kg of Fluoride ion; as Fluoride ion</p> <p>Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient; in preparations for injection except in preparations containing ≤0.1% of Iron</p> <p>Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient; in other preparations except in liquid or gel preparations containing ≤0.1% of Iron, or in animal feeds or feed premixes</p> <p>Schedule 5 listed - for use as agricultural chemicals except in preparations containing ≤4% of Iron</p> <p>Schedule 6 listed - except: when included in Schedule 5, in preparations for human use, or in preparations containing ≤15 mg/kg of Fluoride ion</p> <p>Schedule 6 listed - except up to 1% of Iron oxides when present as an excipient. For the treatment of animals except: when included in Schedule 5, in liquid or gel preparations containing ≤0.1% of Iron, or in animal feeds or feed premixes</p> |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory**

Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component       | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Ferric fluoride | -    | -     | 232-002-4 | -      | X    | X   | -    | X     | X    | X    | -     | KE-21135 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations****Ozone Depletion Potential**

This product does not contain any known or suspected substance

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)**

Not applicable



**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component                     | CAS No     | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------------------|------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Iron(III) fluoride trihydrate | 15469-38-2 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Ferric fluoride               | 7783-50-8  | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

Not applicable

**Section 16 - Other Information****Legend**

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration (PNEC)  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail  
**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Revision Date**

18-Nov-2022

**Revision Summary**

SDS sections updated, 3, 8, 11, 12, 15.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**